MATH 2059 - FINAL EXAM

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~~Hmmm~~

Q1)

Component	Copper	Zinc	Glass	
Transistors	4	1	2	→ x
Resistors	3	3	1	→ y
Computer chips	2	1	3	→ z

$$4x + 3y + 2z \leq 960$$
$$x + 3y + z \leq 510$$
$$2x + y + 3z \leq 610$$

$$\left[\begin{array}{ccc|c} 4 & 3 & 2 & 960 \\ 1 & 3 & 1 & 510 \\ 2 & 1 & 3 & 610 \end{array}\right]$$

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Q2)

t, min	1	2	3.25	4.5	6	7	8	9	9.5	10
v, min	5 y_0	6 y_1	5,5 y_2	7 y_3	8,5 y_4	8 y_5	6 y_6	7 y_7	7 y_8	5 y_9

a) Trapezoidal rule;

$h = \frac{b-a}{n}$ $\quad a=1 \quad b=10 \quad n=9$

$$\int v(t)\,dt = \frac{h}{2}\left[(y_0+y_9) + 2(y_1+y_2+y_3+y_4+y_5+y_6+y_7+y_8)\right]$$

$$= \frac{1}{2}\left[(5+5) + 2(6+5,5+7+8,5+8+6+7+7)\right]$$

$$= \frac{1}{2}\underbrace{\left[10+2(55)\right]}_{120} \rightarrow \frac{120}{2} = 60$$

b) Best combination is Simpson's Rules

$$\int v(t)\,dt = \frac{h}{3}\left[\underbrace{(y_0+y_9)}_{10} + 4\underbrace{(y_1+y_3+y_5+y_7)}_{28} + 2\underbrace{(y_2+y_4+y_6+y_8)}_{27}\right]$$

$$= \frac{1}{3}\underbrace{\left[10+(4\cdot 28)+(2\cdot 27)\right]}_{176}$$

$$= \frac{176}{3} = 58,6$$

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3)

t	0	0.1	0.2	0.3	0.5	0.7
i	0	0.16	0.32	0.56	0.84	2.0

$$V_L = L\frac{di}{dt}$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$$

t=0 $\quad x_i = 0 \qquad x_{i+1} = 0.1 \qquad x_{i+2} = 0.2$

$$f'(0) = \frac{-f(0.2) + 4f(0.1) - 3f(0)}{2(0.1)} \rightarrow \frac{di}{dt} = \frac{-0.32 + 4(0.16) - 3(0)}{2(0.1)} = 1.6$$

t=0.1 $\quad x_i = 0.1 \qquad x_{i+1} = 0.2 \qquad x_{i-1} = 0$

$$f'(0.1) = \frac{f(0.2) - f(0)}{2(0.1)} \rightarrow \frac{di}{dt} = \frac{0.32 - 0}{2(0.1)} = 1.6$$

t=0.2 $\quad x_i = 0.2 \qquad x_{i+1} = 0.3 \qquad x_{i-1} = 0.1$

$$f'(0.2) = \frac{f(0.3) - f(0.1)}{2(0.1)} \rightarrow \frac{di}{dt} = \frac{0.56 - 0.16}{2(0.1)} = 2$$

t=0.3

$$f'(x_i) = f(x_{i-1})\frac{2x_i - x_i - x_{i+1}}{(x_{i-1} - x_i)(x_{i-1} - x_{i+1})} + f(x_i)\frac{2x_i - (x_{i-1}) - x_{i+1}}{(x_i - x_{i-1})(x_i - y_{i+1})} + f(x_{i+1})\frac{2x_i - x_{i-1} - x_i}{(x_{i+1} - x_{i-1})(x_{i+1} - x_i)}$$

↳ $x_i = 0.3 \qquad x_{i+1} = 0.5 \qquad x_{i-1} = 0.2$

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$$f'(0,3) = f(0,2)\,\frac{2(0,3)-(0,3)-(0,5)}{(0,2-0,3)(0,2-0,5)} + f(0,3)\,\frac{2(0,3)-(0,2)-(0,5)}{(0,3-0,2)(0,3-0,5)} + f(0,5)\,\frac{2(0,3)-(0,2)-(0,3)}{(0,5-0,2)(0,5-0,3)}$$

$f(0,2) \to 0,32$; $f(0,3) \to 0,56$; $f(0,5) \to 0,84$

$$\frac{di}{dt} = \boxed{2,066}$$

t = 0,5

$$f'(x_i^o) = \frac{f(x_{i+1}^o) - f(x_{i-1}^o)}{2h}$$

$x_i^o = 0,5$ $\quad x_{i+1}^o = 0,7$ $\quad x_{i-1}^o = 0,3$

$$f'(0,5) = \frac{f(0,7) - f(0,3)}{2\cdot(0,2)} \qquad \frac{di}{dt} = \frac{2-0,56}{2(0,2)} = \underline{3,6}$$

t = 0,7

$$f'(x_i^o) = \frac{f(x_{i-2}^o) - 4f(x_{i-1}^o) + 3f(x_i^o)}{2h}$$

$x_i^o = 0,7$ $\quad x_{i-1}^o = 0,5$ $\quad x_{i-2}^o = 0,3$

$$f'(0,7) = \frac{f(0,3) - 4f(0,5) + 3f(0,7)}{2(0,2)}$$

$$\frac{di}{dt} = \frac{0,56 - 4(0,84) + 3(2)}{0,4} = \underline{8}$$

t	i	(H) $\frac{di}{dt}$	(4H) $V_L = L\frac{di}{dt}$
0	0	1,6	6,4
0,1	0,16	1,6	6,4
0,2	0,32	2	8
0,3	0,56	2,066	8,26
0,5	0,84	3,6	14,4
0,7	2	8	32

[redacted]

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